THE MASTERMIND COACHING CLASSES

Revision Notes

Class 8 · Mathematics

Ganita Prakash — Part 1, Chapters 1 to 7 · CBSE / RBSE Syllabus

HOW TO USE THESE NOTES

Each chapter follows the textbook's own order of sections, so you can read straight through. Bold words are keywords you must be able to define. Tables compare and classify; shaded boxes hold definitions, formulas and key facts. The navy Key Takeaways box at the end of every chapter is your last-two-minutes revision.

CH.	CHAPTER TITLE	CORE THEME
1	A Square and a Cube	Squares, square roots, Pythagorean triplets, cubes, cube roots, Hardy–Ramanujan number
2	Power Play	Exponents, laws of exponents, zero and negative exponents, powers of 10, standard form
3	A Story of Numbers	Counting, tally and one-to-one mapping, Egyptian, Roman, Mesopotamian and Hindu number systems, place value, bases
4	Quadrilaterals	Angle sum, types of quadrilaterals, properties of parallelogram, rhombus, rectangle, square, kite and trapezium
5	Number Play	Divisibility tests with reasons, digit puzzles, generalising with letters, LCM and remainder problems
6	We Distribute, Yet Things Multiply	Distributive property, multiplying expressions, algebraic identities, quick arithmetic, common mistakes
7	Proportional Reasoning – 1	Ratio, equivalent ratios, proportion, unitary method, direct proportion, percentages and change

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CHAPTER 01 · MATHS (PART 1)

A Square and a Cube

Squares and cubes come from the area of a square and the volume of a cube. This chapter studies perfect squares and cubes, their patterns and properties, how to find square roots and cube roots by prime factorisation, how to make a number a perfect square or cube, and Pythagorean triplets.

Big Questions

- What patterns do square numbers and cubes follow?
- How do we find square roots and cube roots using prime factorisation?
- How are Pythagorean triplets formed and what is special about 1729?

1.1 Square Numbers

DEFINITION

A **perfect square** is the square of a natural number: $n \times n = n^2$. Its **square root** $\sqrt{(n^2)} = n$. Geometrically n^2 is the area of a square of side n.

Property	Statement	Example
Last digit	A perfect square ends in **0, 1, 4, 5, 6 or 9** (never 2, 3, 7, 8)	576 ends in 6; 1,022 cannot be a square
Zeros	Number of end zeros is **even**	$400 = 20^2$, $4{,}900 = 70^2$
Parity	Square of an even number is even; of an odd number is odd	$12^2 = 144$; $13^2 = 169$
Between squares	There are **2n** numbers between n^2 and $(n + 1)^2$	between 15^2 and 16^2: 30 numbers
Odd-number sum	$n^2 = 1 + 3 + 5 + \ldots$ (n odd numbers)	$1 + 3 + 5 + 7 + 9 = 25$
Consecutive squares	$(n + 1)^2 - n^2 = 2n + 1$	$26^2 - 25^2 = 51$
Triangular numbers	Sum of two consecutive triangular numbers is a square	$10 + 15 = 25$

FORMULA

Numbers ending in 5: $(10a + 5)^2 = a(a + 1)$ hundred + 25. So $35^2 = (3 \times 4) \mid 25 = 1225$; $105^2 = (10 \times 11) \mid 25 = 11{,}025$. **Consecutive numbers:** $n^2 = (n - 1)(n + 1) + 1$.

1.2 Square Roots by Prime Factorisation

Step 1
Write the prime factorisation of the number.

Step 2
Pair equal primes. A perfect square has **every prime in pairs**.

Step 3
Take **one prime from each pair** and multiply them: that is the square root.

Example. $3136 = 2^6 \times 7^2 = (2^3 \times 7)^2$, so $\sqrt{3136} = 8 \times 7 = \mathbf{56}$. And $576 = 2^6 \times 3^2$ gives $\sqrt{576} = 8 \times 3 = \mathbf{24}$.

▸ Making a perfect square

$180 = 2^2 \times 3^2 \times 5$. The prime **5** is unpaired. Multiply by **5**: $900 = 30^2$. Divide instead: $180 \div 5 = 36 = 6^2$.

Number	Unpaired prime(s)	Smallest multiplier	Square root of the result
180	5	5	30
50	2	2	10
108	3	3	18
294	6 (2 × 3)	6	42

1.3 Pythagorean Triplets

DEFINITION

Three natural numbers (a, b, c) with $\mathbf{a^2 + b^2 = c^2}$ form a **Pythagorean triplet**: they are the sides of a right triangle.

Rule	Triplet	Example
For any natural m > 1: $(2m, m^2 - 1, m^2 + 1)$	m = 2: (4, 3, 5); m = 3: (6, 8, 10); m = 4: (8, 15, 17)	All members from this rule satisfy $a^2 + b^2 = c^2$
For odd m: $(m, (m^2 - 1)/2, (m^2 + 1)/2)$	m = 9: (9, 40, 41)	Triplet whose smallest member is 9

Verify (9, 40, 41): $81 + 1600 = 1681 = 41^2$. Multiples of a triplet are triplets: (3, 4, 5) → (6, 8, 10) → (9, 12, 15).

1.4 Cubes and Cube Roots

DEFINITION

A **perfect cube** is $n \times n \times n = n^3$ (the volume of a cube of side n). Its **cube root** $\sqrt[3]{(n^3)} = n$. Cubes: 1, 8, 27, 64, 125, 216, 343, 512, 729, 1000.

- Cubes of **odd** numbers are odd and of **even** numbers are even; cubes can end in any digit 0–9.
- **Sum of consecutive odd numbers:** $1^3 = 1$, $2^3 = 3 + 5$, $3^3 = 7 + 9 + 11$, $4^3 = 13 + 15 + 17 + 19$.
- **Prime factorisation test:** a number is a perfect cube if every prime occurs in **groups of three**.

Cube root by prime factorisation. $2744 = 2^3 \times 7^3$, so $\sqrt[3]{2744} = 2 \times 7 = \mathbf{14}$. $10648 = 2^3 \times 11^3$ gives **22**. For $1372 = 2^2 \times 7^3$ the prime 2 appears twice, so 1372 is **not** a cube; multiply by 2 to get $2744 = 14^3$.

REMEMBER

To make a number a perfect cube, **multiply** it by the smallest number that completes every prime to a group of three (or divide to remove incomplete groups).

▸ The Hardy–Ramanujan number 1729

1729 is the smallest number that is the sum of two cubes in **two different ways**: $1729 = 1^3 + 12^3 = 9^3 + 10^3$. Hardy remembered the number of his taxi as ‘dull’; Ramanujan at once saw this property.

Quick Revision: Likely Exam Questions

Likely question	Model answer
Is 576 a perfect square? Give √576.	Yes; 576 = 24 × 24, so √576 = 24.
Sum of the first 5 odd numbers without adding?	$5^2 = 25$.
How many numbers lie between 15^2 and 16^2?	2 × 15 = 30.
Square 105 using the ending-5 pattern.	(10 × 11) | 25 = 11,025.
Cube root of 2744 by prime factorisation.	$2^3 \times 7^3$ gives 14.
Smallest number to multiply 180 to make a perfect square?	5; the square root of 900 is 30.
Find a Pythagorean triplet with smallest member 9.	(9, 40, 41).
Is 1372 a perfect cube? What to multiply?	No; $1372 = 2^2 \times 7^3$, multiply by 2.
Show that two consecutive triangular numbers add to a square.	$10 + 15 = 25 = 5^2$; in general $T(n) + T(n + 1) = (n + 1)^2$.
What is special about 1729?	Smallest number expressible as a sum of two cubes in two ways: $1^3 + 12^3$ and $9^3 + 10^3$.

★ KEY TAKEAWAYS

- A **perfect square** ends in 0, 1, 4, 5, 6 or 9; between n^2 and $(n + 1)^2$ there are **2n** numbers.
- **Square root / cube root:** group the prime factors in **pairs / triples**.
- **$(2m, m^2 – 1, m^2 + 1)$** gives Pythagorean triplets; check with $a^2 + b^2 = c^2$.
- n^2 is the sum of the first **n odd numbers**; n^3 is the sum of n **consecutive odd numbers**.
- **$1729 = 1^3 + 12^3 = 9^3 + 10^3$**, the Hardy–Ramanujan number.

CHAPTER 02 · MATHS (PART 1)

Power Play

Repeated multiplication is written compactly with powers. This chapter defines exponents, develops the laws of exponents, extends them to zero and negative exponents, expresses numbers as powers of 10 and writes very large and very small numbers in standard (scientific) form.

Big Questions

- What do an exponent and a base mean and how do the laws of exponents follow?
- What do zero and negative exponents mean?
- How are huge and tiny numbers written and compared in standard form?

2.1 Powers and Exponents

DEFINITION

a^n = a × a × … × a (n times). Here **a is the base** and **n is the exponent (power)**. $2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32$. A power grows very fast: $2^{10} = 1{,}024$ is far larger than $10^2 = 100$.

- $(-2)^4 = 16$ but $(-2)^3 = -8$: an **even** power of a negative number is positive, an **odd** power is negative.
- Order matters: $3^2 = 9$ but $2^3 = 8$; $2^5 = 32$ but $5^2 = 25$.
- **Powers of 10:** $10^2 = 100$, $10^3 = 1{,}000$, $10^6 = 10$ lakh, $10^7 = 1$ crore. The exponent = **number of zeros**.

2.2 Laws of Exponents

Law	Rule (a, b ≠ 0)	Example
Product	$a^m \times a^n = a^{m+n}$	$2^3 \times 2^4 = 2^7 = 128$
Quotient	$a^m \div a^n = a^{m-n}$	$5^6 \div 5^2 = 5^4 = 625$
Power of a power	$(a^m)^n = a^{mn}$	$(3^2)^3 = 3^6 = 729$
Product to a power	$a^m \times b^m = (ab)^m$	$2^3 \times 5^3 = 10^3 = 1{,}000$
Quotient to a power	$a^m \div b^m = (a/b)^m$	$6^4 \div 3^4 = 2^4 = 16$

IMPORTANT

The product and quotient laws need the **same base**. $2^3 \times 3^2$ cannot be combined as a single power. Also $a^m + a^n$ has **no** simple law: $2^3 + 2^2 = 12$, not 2^5.

2.3 Zero and Negative Exponents

FORMULA

$a^0 = 1$ ($a \neq 0$). **$a^{-n} = 1/a^n$**, the **reciprocal** of a^n. Also $(a/b)^{-n} = (b/a)^n$.

Why $a^0 = 1$: continue the pattern $2^3 = 8$, $2^2 = 4$, $2^1 = 2$ (each time dividing by 2): the next is $2^0 = 1$, then $2^{-1} = 1/2$, $2^{-2} = 1/4$. By the quotient law, $5^3 \div 5^3 = 5^0 = 1$.

Expression	Value
$4^0 + 4^{-1}$	$1 + 1/4 = 5/4$
$2^5 \times 2^{-3}$	$2^2 = 4$
$(3/4)^{-2}$	$(4/3)^2 = 16/9$
10^{-3}	$1/1000 = 0.001$
$(10^5 \times 10^{-2}) \div 10^2$	$10^1 = 10$

2.4 Standard (Scientific) Form

DEFINITION

A number is in **standard form** when written as $\mathbf{k \times 10^n}$ with $\mathbf{1 \le k < 10}$ and n an integer. Large numbers have **positive** n; numbers smaller than 1 have **negative** n.

Large number

Move the decimal point left until one non-zero digit is before it; n = number of places moved. 62,00,00,000 = 6.2×10^8.

Small number

Move the decimal point right; n is negative. 0.00045 = 4.5×10^{-4}.

Back to usual form

6.2×10^5 = 6,20,000; $3 \times 10^{-2} = 0.03$.

Quantity	Standard form
Mass of Earth	5.97×10^{24} kg
Mass of Moon	7.35×10^{22} kg
Speed of light	3×10^8 m/s
Diameter of a hydrogen atom (approx.)	1×10^{-10} m

Comparing: first compare the exponents of 10; the larger exponent is the larger number. $2 \times 10^5 > 3 \times 10^4$ and Earth's mass is about $5.97 \times 10^{24} \div 7.35 \times 10^{22} \approx 81$ times the Moon's. **Dividing:** $(4.2 \times 10^5) \div (2.1 \times 10^2) = 2 \times 10^3$.

REMEMBER

To add or subtract numbers in standard form, **first make the powers of 10 equal**: $3 \times 10^4 + 2 \times 10^3 = 30 \times 10^3 + 2 \times 10^3 = 32 \times 10^3 = 3.2 \times 10^4$.

Quick Revision: Likely Exam Questions

Likely question	Model answer
Simplify $2^5 \times 2^{-3}$.	$2^2 = 4$.
Write 6.2×10^5 in usual form.	6,20,000.
Find $4^0 + 4^{-1}$.	5/4.
Which is greater, 2^{10} or 10^2?	$2^{10} = 1{,}024 > 100$.
Simplify $(10^5 \times 10^{-2}) \div 10^2$.	10.
Compare 3×10^4 and 2×10^5.	2×10^5 is larger.
Simplify $(3/4)^{-2}$.	16/9.
State the law for $(a^m)^n$.	a^{mn}.
Write 0.00045 in standard form.	4.5×10^{-4}.
Why is $a^0 = 1$?	From $a^m \div a^m = a^{m-m} = a^0$ and the quotient equals 1.

★ KEY TAKEAWAYS

- **a^n** means n factors of a; sign of $(-a)^n$ depends on whether n is even or odd.
- Laws: **$a^m \times a^n = a^{m+n}$, $a^m \div a^n = a^{m-n}$, $(a^m)^n = a^{mn}$, $a^m b^m = (ab)^m$.**
- **$a^0 = 1$** and **$a^{-n} = 1/a^n$**.
- **Standard form:** $k \times 10^n$, $1 \le k < 10$; negative n for numbers less than 1.
- Compare standard-form numbers by the **power of 10 first**, then by k.

CHAPTER 03 · MATHS (PART 1)

A Story of Numbers

Humans have counted for tens of thousands of years. This chapter traces how counting grew from tally marks and one-to-one matching to the number systems of Egypt, Rome and Mesopotamia and finally to the Hindu place-value system with zero, and shows how place value works in other bases such as 5 and 8.

Big Questions

- How did early people count and record numbers?
- How do additive systems (Egyptian, Roman) differ from place-value systems?
- Why is the Hindu place-value system with zero so powerful and how do other bases work?

3.1 Early Counting

- **One-to-one mapping:** pairing each object with a pebble, notch or knot (a shepherd matching sheep with stones) without needing number names.
- **Tally marks:** a stroke for each object, grouped in **fives** so that large counts are easy to read. The Ishango bone (about 20,000 years old) has such notches.
- **Limits:** tally marks are long for big numbers, so people invented **symbols** for groups (tens, hundreds, …).

3.2 Additive Systems

System	How numbers are written	Features
Egyptian (hieroglyphs)	separate symbols for 1, 10, 100, 1000, …, repeated as needed	base 10, additive, no zero, no place value
Roman	I = 1, V = 5, X = 10, L = 50, C = 100, D = 500, M = 1000	additive and subtractive (IV = 4, IX = 9, XL = 40, XC = 90, CD = 400, CM = 900); no zero

Roman examples. 2499 = MM CD XC IX = **MMCDXCIX**; 949 = CM XL IX = **CMXLIX**. MCDXL + DLX = 1440 + 560 = 2000 = **MM**. The Roman system is **not suitable for big calculations**: there is no zero, no place value, and long strings of symbols are needed.

> **REMEMBER**
> Roman rules: a symbol may be repeated up to **three** times (III = 3); V, L, D are never repeated or subtracted; a smaller symbol placed **before** a larger one is subtracted.

3.3 The Mesopotamian System

The Babylonians (Mesopotamia) wrote on clay tablets with a wedge-shaped stylus. They used **base 60** with two symbols, a vertical wedge for **1** and a corner wedge for **10**. Value depended on **position**, so it was a place-value system, but early versions had no symbol for an empty place. Our 60 minutes in an hour and 360° in a circle come from base 60.

3.4 The Hindu (Indian) Place-Value System

DEFINITION

In a **place-value system** the value of a digit depends on its **position**. In base 10, 734 = 7 × 100 + 3 × 10 + 4 × 1. The Hindu system uses **ten symbols (0–9)**, with **zero as a placeholder and a number**.

Contribution	Why it matters
Ten symbols 0–9 (Brahmi numerals developed into modern digits)	any number can be written using only 10 symbols
Zero as a placeholder and as a number (Brahmagupta's rules, 7th century)	separates 305 from 35 and allows arithmetic with zero
Place value	short, easy addition and multiplication
Transmission to the world	spread through Arab scholars to Europe as the **Hindu–Arabic numerals**

Indian scholars such as **Āryabhaṭa** (5th century) used place value in calculation, and **Brahmagupta** gave the rules for addition, subtraction and multiplication with zero and negative numbers.

3.5 Other Bases

FORMULA

In **base b** the place values from the right are 1, b, b^2, b^3, … So a 3-digit number **pqr in base b = p × b^2 + q × b + r**.

Base-n number	Working	Decimal value
734 in base 8 (written 734_8)	7 × 64 + 3 × 8 + 4	476
1234 in base 5	1 × 125 + 2 × 25 + 3 × 5 + 4	194
1031 in base 4	1 × 64 + 0 × 16 + 3 × 4 + 1	77

Convert 88 to base 5

88 ÷ 5 = 17 remainder 3; 17 ÷ 5 = 3 remainder 2; 3 ÷ 5 = 0 remainder 3.

Read the remainders upward

88 = 323 in base 5 (3 × 25 + 2 × 5 + 3 = 88).

Convert 77 to base 4

77 ÷ 4 = 19 r 1; 19 ÷ 4 = 4 r 3; 4 ÷ 4 = 1 r 0; 1 ÷ 4 = 0 r 1: 77 = 1031 in base 4.

NOTE

Computers work in **base 2 (binary)** with the digits 0 and 1: 13 = 1101 in base 2 because 8 + 4 + 0 + 1 = 13.

Quick Revision: Likely Exam Questions

Likely question	Model answer
What is one-to-one mapping?	Pairing each object with exactly one token or mark, e.g. a stone for each cow.
Why is the Roman system unsuitable for large calculations?	No zero or place value and long strings of symbols.
Why is the Hindu number system a great invention?	Ten symbols and place value (with zero) express every number and make calculation simple.
What is the role of zero?	It holds an empty place so 305 differs from 35, and it is a number itself.
Write 949 in Roman numerals.	CMXLIX.
Convert 88 to base 5.	323.
Convert 1234 in base 5 to decimal.	194.
Place values in 734 base 8?	7 × 64, 3 × 8, 4 × 1.
How did the Mesopotamians write numbers?	Base 60 using wedge symbols for 1 and 10, place-value positions.
What is a tally mark system?	Counting by one stroke per object, grouped in fives.

★ KEY TAKEAWAYS

- Counting began with **one-to-one mapping** and **tally marks**.
- **Egyptian and Roman** systems are additive and have **no zero** or place value.
- **Mesopotamians** used base 60 with place value; our time and angle units come from it.
- The **Hindu system** uses ten digits, **place value and zero**, and spread to the whole world.
- In **base b**, place values are 1, b, b^2, …; convert by repeated division.

CHAPTER 04 · MATHS (PART 1)

Quadrilaterals

A quadrilateral is a closed figure with four sides. This chapter proves that its angles add to 360°, classifies the special quadrilaterals by sides, angles and diagonals, shows how they are related to one another and uses their properties to find unknown angles and lengths.

Big Questions

- What is the angle sum of a quadrilateral and why?
- Which properties define a parallelogram, rhombus, rectangle, square, kite and trapezium?
- How are these shapes related and how do we use their properties to solve problems?

4.1 Angle Sum Property

FORMULA

The four interior angles of any quadrilateral add up to **360°**. **Reason:** a diagonal splits it into **two triangles**, each with angle sum 180°, so 2 × 180° = 360°.

Problem	Working	Answer
Angles in the ratio 3 : 4 : 5 : 6	18x = 360, x = 20	60°, 80°, 100°, 120°
Angles in the ratio 1 : 3 : 4 : 7	15x = 360, x = 24	24°, 72°, 96°, 168°
x, x + 20, x + 40, x + 60	4x + 120 = 360	x = 60; angles 60°, 80°, 100°, 120°

4.2 The Special Quadrilaterals

Shape	Sides	Angles	Diagonals
Trapezium	exactly one pair of parallel sides	angles on the same leg add to 180°	no special property (equal in an isosceles trapezium)
Parallelogram	both pairs of opposite sides parallel and **equal**	opposite angles **equal**; adjacent angles **supplementary** (180°)	**bisect each other**
Rectangle	opposite sides equal and parallel	all angles **90°**	**equal** and bisect each other
Rhombus	**all four sides equal**; opposite sides parallel	opposite angles equal	bisect each other **at right angles**; bisect the angles
Square	all sides equal	all angles 90°	equal, bisect each other at right angles
Kite	two pairs of **adjacent** sides equal	one pair of opposite angles equal	perpendicular; one diagonal bisects the other

▸ How the shapes are related

- Every **square** is a rectangle, a rhombus and a parallelogram.

- Every **rectangle** and every **rhombus** is a parallelogram.
- If a trapezium is defined as having **at least** one pair of parallel sides, every parallelogram is also a trapezium; with the ‘exactly one pair’ definition it is not. Follow your textbook's definition.
- A **rhombus is a kite** with all four sides equal.

> **IMPORTANT**
> A square is a **special rhombus** (it also has right angles) and a **special rectangle** (it also has equal sides). A rectangle is **not** a rhombus unless all its sides are equal.

4.3 Using the Properties

▸ Parallelogram

In a parallelogram with $\angle A = 70°$: $\angle C = 70°$ (opposite), $\angle B = \angle D = 180° - 70° =$ **110°**. If $\angle A = (3x + 10)°$ and $\angle B = (2x - 10)°$, then $5x = 180$, $x = 36$ and the angles are **118° and 62°**.

▸ Rhombus

Diagonals 16 cm and 30 cm bisect each other at 90°, so each side is the hypotenuse of a right triangle with legs 8 and 15: side = $\sqrt{(64 + 225)} = \sqrt{289} =$ **17 cm**; perimeter $4 \times 17 =$ **68 cm**. Area of a rhombus = $\frac{1}{2} \times d_1 \times d_2 = \frac{1}{2} \times 16 \times 30 =$ **240 cm^2**.

▸ Kite

In kite ABCD with AB = AD, CB = CD, $\angle B = \angle D = 90°$ and $\angle A = 70°$: $\angle C = 360° - (90° + 90° + 70°) =$ **110°**. The diagonal AC is the **axis of symmetry** and is the perpendicular bisector of BD.

▸ Trapezium

> **FORMULA**
> **Area of a trapezium = ½ × (sum of parallel sides) × height.** Parallel sides 10 cm and 6 cm, height 4 cm: $\frac{1}{2} \times 16 \times 4 =$ **32 cm^2**.

4.4 Proving a Quadrilateral Is Special

If we know …	… then the quadrilateral is a
both pairs of opposite sides equal	parallelogram
diagonals bisect each other	parallelogram
diagonals bisect each other and are equal	rectangle
diagonals bisect each other at 90°	rhombus
diagonals equal, bisect each other at 90°	square
one pair of sides both equal and parallel	parallelogram

Quick Revision: Likely Exam Questions

Likely question	Model answer
State the angle sum property of a quadrilateral.	Interior angles add up to 360°.
Two properties of the diagonals of a rhombus.	They bisect each other at right angles and bisect the angles.
One angle of a parallelogram is 70°. Find the other three.	70°, 110°, 110°.
Sum of adjacent angles of a parallelogram?	180°.
Quadrilateral with equal diagonals that are not perpendicular?	A rectangle.
Why is a square a special rhombus?	It has all sides equal and also right angles.
Angles in the ratio 3 : 4 : 5 : 6. Find them.	60°, 80°, 100°, 120°.
Diagonals of a rhombus are 16 and 30 cm. Find its side.	17 cm.
Area of a trapezium with parallel sides 10 and 6 cm, height 4 cm?	32 cm^2.
In a kite ∠B = ∠D = 90°, ∠A = 70°. Find ∠C.	110°.

★ KEY TAKEAWAYS

- Angle sum of a quadrilateral = **360°** (two triangles).
- **Parallelogram:** opposite sides and angles equal, diagonals bisect each other.
- **Rectangle:** equal diagonals; **rhombus:** perpendicular diagonals; **square:** both.
- A **kite** has two pairs of equal adjacent sides; a **trapezium** has one pair of parallel sides.
- Square ⊂ rectangle, rhombus ⊂ parallelogram; trapezium area = **½ (a + b) h**.

CHAPTER 05 · MATHS (PART 1)

Number Play

This chapter goes deeper into numbers than earlier years. You learn why the divisibility tests work, use letters to describe digits and prove number tricks for every case, solve missing-digit puzzles, and use LCM and primes to answer questions about remainders and common multiples.

Big Questions

- Why do the divisibility tests for 3, 9, 11 and 8 work?
- How can we describe numbers by their digits using letters and prove tricks in general?
- How do LCM and remainders solve puzzles about repeated events?

5.1 Divisibility Tests and Their Reasons

By	Test	Reason (idea)	Example
2, 5, 10	last digit	10 is a multiple of 2, 5 and 10, so only the ones digit matters	7,390 ✓ by 2, 5, 10
4	last two digits	100 is a multiple of 4	5,316: 16 ✓
8	last three digits	1000 is a multiple of 8	7,392: 392 = 8 × 49 ✓
3	digit sum divisible by 3	10, 100, 1000, … each leave remainder 1 on division by 3 (and by 9)	2,184: 15 ✓
9	digit sum divisible by 9	same reason	8,127: 18 ✓
11	(sum of digits at odd places) − (sum at even places) is 0 or a multiple of 11	10, 1000, … leave remainder −1; 100, 10000, … leave +1	3,652: (3 + 5) − (6 + 2) = 0 ✓

IMPORTANT

For a **composite** divisor use **co-prime** factors: divisible by 12 ⇔ divisible by **3 and 4**; by 6 ⇔ by 2 and 3; by 15 ⇔ by 3 and 5. Divisibility by 2 and 4 does **not** guarantee divisibility by 8.

5.2 Missing-Digit Puzzles

Choose the test

Pick the rule for the required divisor.

Form the condition

8□16 divisible by 9: digit sum 8 + □ + 1 + 6 = 15 + □ must be a multiple of 9, so □ = **3**.

List all answers

9□36 divisible by 3: 18 + □ divisible by 3, so □ = 0, 3, 6 or 9.

Check

3□52 divisible by 11: (3 + 5) − (□ + 2) = 6 − □ must be 0 or ±11, so □ = **6**.

5.3 Describing Numbers with Letters

FORMULA

A 2-digit number with tens digit a and ones digit b is **10a + b**; its reverse is **10b + a**. A 3-digit number is **100a + 10b + c**.

Trick	Proof	Result
Number + its reverse	(10a + b) + (10b + a) = 11(a + b)	always a **multiple of 11**
Number − its reverse	(10a + b) − (10b + a) = 9(a − b)	always a **multiple of 9**
Digit sum of a 2-digit number is 9	a + b = 9; reverse − number = 9(b − a) = 9 × (9 − 2a)	reversed number differs by a multiple of 9
Sum of 3 consecutive numbers	n + (n + 1) + (n + 2) = 3(n + 1)	multiple of **3**

Using letters shows a result is true for **every** number. A single numerical example only illustrates it and can never prove it.

5.4 LCM, Primes and Remainders

- **LCM of 6, 8, 12:** 6 = 2 × 3, 8 = 2^3, 12 = 2^2 × 3, so LCM = 2^3 × 3 = **24**. **Smallest number divisible by 15, 20 and 25:** 15 = 3 × 5, 20 = 2^2 × 5, 25 = 5^2, LCM = 2^2 × 3 × 5^2 = **300**.
- **Same remainder:** a number leaving remainder 3 when divided by 5, 6 and 8 is of the form LCM(5, 6, 8) × k + 3 = 120k + 3; the smallest such number greater than 3 is **123**.
- **Repeating events:** bells ringing every 12, 18 and 24 minutes ring together after LCM = **72 minutes**.
- **2 is the only even prime**, since every other even number has 2 as an extra factor. Every prime p > 3 has the form 6k ± 1.
- The sum of the first 10 odd numbers is 10^2 = **100**.

REMEMBER

To find the **remainder pattern**, subtract the remainder first: if N leaves 3 on division by 5, 6 and 8 then **N − 3** is a common multiple of 5, 6, 8.

Quick Revision: Likely Exam Questions

Likely question	Model answer
State the divisibility test for 3.	A number is divisible by 3 if its digit sum is divisible by 3.
Is 7392 divisible by 8?	Yes; 392 = 8 × 49.
LCM of 6, 8 and 12?	24.
Why is 2 the only even prime?	Every other even number has 2 as a factor besides 1 and itself.
Is a number divisible by 12 if it is divisible by 3 and 4?	Yes; 3 and 4 are co-prime and 3 × 4 = 12.
Digit for 8□16 divisible by 9?	3.

Likely question	Model answer
Show that a 2-digit number plus its reverse is a multiple of 11.	(10a + b) + (10b + a) = 11(a + b).
Smallest number divisible by 15, 20 and 25?	300.
Smallest number leaving remainder 3 on division by 5, 6, 8 (greater than 3)?	123.
Sum of the first 10 odd numbers?	$100 = 10^2$.

★ KEY TAKEAWAYS

- Tests for **2, 4, 8, 5, 10** look at the last 1, 2, 3 digits; for **3, 9** at the digit sum; for **11** at alternate digit sums.
- For composite divisors test **co-prime factors**.
- A 2-digit number is **10a + b**; use letters to **prove** tricks for all cases.
- Number + reverse = **11(a + b)**; number − reverse = **9(a − b)**.
- **LCM** answers common-multiple and remainder problems: N = LCM × k + r.

CHAPTER 06 · MATHS (PART 1)

We Distribute, Yet Things Multiply

The distributive property says that multiplying a sum by a number is the same as multiplying each part and adding. This chapter extends it to products of two expressions, derives the standard identities for squares and for the difference of squares, and uses them to calculate quickly and to avoid common algebra mistakes.

Big Questions

- How does the distributive property work for numbers and for algebraic expressions?
- What are the identities for $(a + b)^2$, $(a - b)^2$, $(a + b)(a - b)$ and $(x + a)(x + b)$?
- How do identities speed up arithmetic and how do we spot common mistakes?

6.1 The Distributive Property

FORMULA

$a \times (b + c) = a \times b + a \times c$ and **$a \times (b - c) = a \times b - a \times c$**. Multiplication distributes over addition and subtraction.

Area explanation: a rectangle of length a and breadth (b + c) can be cut into two rectangles of areas ab and ac. So $7 \times 105 = 7 \times 100 + 7 \times 5 =$ **735**, and $6 \times 98 = 6 \times 100 - 6 \times 2 =$ **588**.

▸ Multiplying two expressions

$(a + b)(c + d) = ac + ad + bc + bd$: **every term of the first bracket multiplies every term of the second**. $(x + 2)(x + 9) = x^2 + 9x + 2x + 18 =$ **$x^2 + 11x + 18$**. Check with x = 1: $3 \times 10 = 30 = 1 + 11 + 18$ ✓. $(3x + 2)(2x - 5) = 6x^2 - 15x + 4x - 10 =$ **$6x^2 - 11x - 10$**.

REMEMBER

Number of terms before simplifying = (terms in the first) × (terms in the second). **Mind the signs**: negative × negative = positive. Verify any expansion by substituting a small number such as x = 1 or 2.

6.2 Standard Identities

DEFINITION

An **identity** is an equality true for **every** value of the letters. An **equation** is true only for particular values.

Identity	Statement	Example
I	**$(a + b)^2 = a^2 + 2ab + b^2$**	$(x + 3)^2 = x^2 + 6x + 9$
II	**$(a - b)^2 = a^2 - 2ab + b^2$**	$(2a - 3)^2 = 4a^2 - 12a + 9$
III	**$(a + b)(a - b) = a^2 - b^2$**	$(x - 4)(x + 4) = x^2 - 16$
IV	**$(x + a)(x + b) = x^2 + (a + b)x + ab$**	$(x + 2)(x + 9) = x^2 + 11x + 18$

Geometric proof of $(a + b)^2$: a square of side $(a + b)$ splits into a square a^2, a square b^2 and two rectangles each ab, giving $a^2 + b^2 + 2ab$. So $(a + b)^2$ is **not** $a^2 + b^2$: for $a = 2$, $b = 3$, $(a + b)^2 = 25$ but $a^2 + b^2 = 13$.

6.3 Using Identities for Quick Calculation

Calculation	Identity used	Working	Answer
19^2	$(20 - 1)^2$	$400 - 40 + 1$	361
102^2	$(100 + 2)^2$	$10{,}000 + 400 + 4$	10,404
998^2	$(1000 - 2)^2$	$1{,}000{,}000 - 4{,}000 + 4$	996,004
$97^2 - 3^2$	$a^2 - b^2 = (a + b)(a - b)$	100×94	9,400
$111^2 - 89^2$	same	$(111 + 89)(111 - 89) = 200 \times 22$	4,400
103×97	$(100 + 3)(100 - 3)$	$10{,}000 - 9$	9,991
52×48	$(50 + 2)(50 - 2)$	$2{,}500 - 4$	2,496

> **NOTE**
>
> **Trick for squares ending in 5:** $(10a + 5)^2 = 100a(a + 1) + 25$, as $75^2 = 7 \times 8 \mid 25 =$ **5625**. It comes from Identity I.

6.4 Mind the Mistake

Wrong working	Why it is wrong	Correct
$(x + 5)^2 = x^2 + 25$	the middle term $2 \times x \times 5$ is missing	$x^2 + 10x + 25$
$(a - b)^2 = a^2 - b^2$	confuses squares with the difference of squares	$a^2 - 2ab + b^2$
$2(x + 3) = 2x + 3$	bracket not fully distributed	$2x + 6$
$-(x - 4) = -x - 4$	sign of the second term not changed	$-x + 4$
$(2a)^2 = 2a^2$	the coefficient is also squared	$4a^2$

Factorising reverses expansion: $x^2 - 16 = (x - 4)(x + 4)$; $4a^2 - 12a + 9 = (2a - 3)^2$; $x^2 + 11x + 18 = (x + 2)(x + 9)$ (find two numbers with sum 11 and product 18).

Quick Revision: Likely Exam Questions

Likely question	Model answer
State the distributive property.	$a(b + c) = ab + ac$.
Expand $(2a - 3)^2$.	$4a^2 - 12a + 9$.
Is $(a + b)^2 = a^2 + b^2$? Verify with $a = 2$, $b = 3$.	No; $25 \neq 13$.
Find $(x - 4)(x + 4)$.	$x^2 - 16$.

Likely question	Model answer
Find 19^2 using $(20 - 1)^2$.	$400 - 40 + 1 = 361$.
Name the identity $(m + n)(m - n) = m^2 - n^2$.	Difference of two squares.
Evaluate $97^2 - 3^2$ without squaring.	$(97 + 3)(97 - 3) = 9{,}400$.
Find 1002^2 using an identity.	$(1000 + 2)^2 = 1{,}004{,}004$.
Multiply $(3x + 2)(2x - 5)$.	$6x^2 - 11x - 10$.
A student writes $(x + 5)^2 = x^2 + 25$. Correct it.	The term $10x$ is missing: $x^2 + 10x + 25$.

★ KEY TAKEAWAYS

- **Distributive property:** $a(b + c) = ab + ac$; multiply **every term by every term**.
- **$(a + b)^2 = a^2 + 2ab + b^2$; $(a - b)^2 = a^2 - 2ab + b^2$; $(a + b)(a - b) = a^2 - b^2$.**
- **$(x + a)(x + b) = x^2 + (a + b)x + ab$.**
- Identities give **quick arithmetic**: $98^2 = (100 - 2)^2$, $103 \times 97 = 100^2 - 3^2$.
- Check expansions by **substituting** a number; avoid the missing middle term mistake.

CHAPTER 07 · MATHS (PART 1)

Proportional Reasoning – 1

Comparing quantities by division, rather than subtraction, is the heart of proportional reasoning. This chapter covers ratios and equivalent ratios, proportion, the unitary method, direct proportion, sharing a quantity in a ratio, and percentages as ratios with the idea of percentage change.

Big Questions

- What is a ratio and when are two ratios in proportion?
- How do we solve problems by the unitary method and by direct proportion?
- How are percentages and percentage changes related to ratios?

7.1 Ratio and Equivalent Ratios

DEFINITION

A **ratio** compares two quantities of the **same kind and unit** by division: a : b = a/b. The ratio 24 : 36 in **simplest form** is **2 : 3** (divide both terms by their HCF 12). Order matters: 2 : 3 is not 3 : 2.

- **Equivalent ratios:** multiply or divide both terms by the same number: 2 : 3 = 4 : 6 = 10 : 15.
- A ratio has **no unit**: 40 cm : 1 m is 40 : 100 = **2 : 5** after converting to the same unit.
- The ratio of the perimeter of a square of side 6 cm to its side is 24 : 6 = **4 : 1**.

7.2 Proportion

DEFINITION

Four numbers a, b, c, d are in **proportion** if a : b = c : d, written a : b :: c : d. Then **a × d = b × c** (product of extremes = product of means).

Finding the missing term: 5 : 8 = x : 40 gives 8x = 200, so x = **25**. Check: 5/8 = 25/40 ✓. In 3 : 5 :: 12 : y, y = 5 × 12 ÷ 3 = **20**.

7.3 Unitary Method and Direct Proportion

DEFINITION

Two quantities are in **direct proportion** if they increase (or decrease) together **in the same ratio**: when one is multiplied by k, the other is multiplied by k. Their ratio y/x stays **constant**.

Unitary method

Find the value of **one unit**, then multiply for the required number. A car uses 8 L of petrol for 112 km: 1 L gives 112 ÷ 8 = 14 km, so 15 L gives 15 × 14 = **210 km**.

Proportion method

Write $x_1 : y_1 = x_2 : y_2$ and solve: 2 cups of rice for 5 persons; for 15 persons 2 : 5 = x : 15, x = **6 cups**.

Direct proportion	Not in direct proportion
Cost and number of items bought	Age and height of a person

Direct proportion	Not in direct proportion
Distance and time at constant speed	Number of workers and days to finish a job (this is **inverse** proportion)
Weight of rice and its price	Marks scored and hours of TV watched

NOTE

If more men do a job in **fewer days**, the quantities are in **inverse proportion**: the product (men × days) is constant. 15 men take 12 days, so 20 men take 15 × 12 ÷ 20 = **9 days**. Inverse proportion is developed in *Proportional Reasoning – 2* of Part 2.

7.4 Sharing in a Given Ratio

Step 1
Add the ratio terms to get the total number of parts.

Step 2
Divide the whole by the total parts: value of one part.

Step 3
Multiply each term by one part.

Check
The shares must add up to the whole.

Problem	Total parts	One part	Shares
₹1,200 in the ratio 3 : 5	8	₹150	₹450 and ₹750
₹4,200 in the ratio 4 : 3	7	₹600	₹2,400 and ₹1,800
₹2,800 in the ratio 3 : 4 : 7	14	₹200	₹600, ₹800, ₹1,400
Two numbers with ratio 4 : 5 and sum 72	9	8	32 and 40

7.5 Percentages and Percentage Change

FORMULA

x% = x/100. Percentage of a quantity = (x/100) × quantity. **Percentage change = (change ÷ original) × 100.**

Fraction	Decimal	Percent
1/2	0.5	50%
1/4	0.25	25%
3/4	0.75	75%
1/5	0.2	20%
1/10	0.1	10%

- **Increase:** a price of ₹800 rises to ₹920: change 120, 120/800 × 100 = **15% increase**.
- **Decrease:** ₹500 falls to ₹400: 100/500 × 100 = **20% decrease**.

- **Caution:** a 20% rise followed by a 20% fall does **not** return to the start: 100 → 120 → 96, a net 4% fall, because the second percentage is taken on a **new** base.
- Ratio 3 : 7 means the first is 3/10 = **30%** of the total and the second 70%.

Quick Revision: Likely Exam Questions

Likely question	Model answer
What is a ratio? Simplify 24 : 36.	A comparison of two like quantities by division; 2 : 3.
Divide ₹1200 between two friends in the ratio 3 : 5.	₹450 and ₹750.
Find x if 5 : 8 = x : 40.	x = 25.
2 cups of rice for 5 persons. How many for 15?	6 cups.
What is direct proportion? Example.	Both quantities increase or decrease in the same ratio, e.g. cost and number of items.
A car uses 8 L for 112 km. Distance for 15 L?	210 km.
Divide ₹2800 in the ratio 3 : 4 : 7.	₹600, ₹800, ₹1400.
Two numbers in the ratio 4 : 5 add to 72. Find them.	32 and 40.
15 men finish a job in 12 days. How long for 20 men?	9 days (inverse proportion).
Price rises from ₹800 to ₹920. Percentage increase?	15%.

★ KEY TAKEAWAYS

- A **ratio** compares like quantities by division; reduce it by the **HCF**.
- In **proportion** a : b = c : d, so **ad = bc**.
- **Direct proportion:** y/x constant; solve by the **unitary method** or by equal ratios.
- **Share** a quantity: total parts → one part → each share; shares add up to the whole.
- **Percentage change = change ÷ original × 100**; successive changes use a new base each time.